Ground Truth README

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1.1 Reference data files description

This section will give a detailed description of the reference data files structure. The reference data reported in this section refers to the Dronetag data recorded per each UAV. Dronetag files are available for each UAV's flight.

1.1.1 Naming conventions

Each sensor data file is a .csv file reporting the naming convention:

DroneTag_<TEAM>_<YYMMDD(takeoff)>_<HHMM(takeoff)>_<HHMM(landing)>_<DroneName>.csv'

Explanation of each entry of the sensor data file name is as follows:

- <TEAM> can either be": 'BLUE' (friendly) or 'RED' (enemy)
- <YYMMDD(takeoff)> - Date (year, month, day [UTC]) at which the drone has took-off.
- <HHMM(takeoff)> - Time (hours and minutes [UTC]) at which the drone has took-off.
- <HHMM(landing)> - Time (hours and minutes [UTC]) at which the drone has landed. The take-off and landing days are the same.
- <DroneName> is the name of the drone

Dronetag data file name example: *"DroneTag_RED_240916_1411_1422_Skydio X10D.csv"*

1.1.2 Data structures

The data in the Dronetag .csv files contain Dronetag report messages. The data in the Sensors .csv files contains detection report messages. These reports contain information about the location of the device, as well as velocity of the UAS carrying the Dronetahd and other device characteristics. These information are provided as comma separated files with self-explanatory column names. Detailed structure description for the dronetag data is reported below.

1.1.2.1 Dronetag

File containing ground-truth data gathered by a single drone during the current scenario and iteration. The corresponding data structure is reported below as well as a brief description of each data entry.

- timestamp [UTC] : The recorded time of the measurement.
- longitude [°]: The geographic longitude coordinate of the device.
- latitude [°]: The geographic latitude coordinate of the device.
- altitude [m]: Altitude calculated from barometric pressure.
- geo_altitude [m]: Altitude above the WGS84 ellipsoid.
- height [m]: Altitude above takeoff point.
- velocity_x [m/s]: Device velocity along the X-axis in ENU.
- velocity_y [m/s]: Device velocity along the Y-axis in ENU.
- velocity_z [m/s]: Device velocity along the Z-axis in ENU.
- heading [°]: Direction the device is facing or moving, measured in degrees from true north (0–360°). Angles increase clockwise (i.e. 90° = East, 180° = South, 270° = West).
- horizontal_accuracy [m]: Estimated accuracy of the latitude/longitude coordinates.
- vertical_accuracy [m]: Estimated accuracy of the altitude measurement.
- speed_accuracy [m/s]: Estimated accuracy of the speed measurement in ENU.
- migrated: Boolean or flag indicating whether the data point has been migrated or post-processed.
- pressure [hPa]: Atmospheric pressure measured by the device's barometer.
- rsrp [dBm]: Reference Signal Received Power of LTE.
- rsrq [dB]: Reference Signal Received Quality of LTE.
- satellites: Number of GNSS satellites used in the position fix.
- snr [dB]: Signal to Noise Ratio of LTE.
- tac: Tracking Area Code of LTE (range from 0 to 65,536)
- battery: Battery level of the device.
- cellid: ID of the current LTE cell (range from 0 to 268,435,455 on UMTS and LTE networks)
- charging: Boolean or flag indicating whether the device is currently charging.
- flight_id: Identifier for the flight track associated with the data point.